

Investigating Whether Inductive Backdoors Are a LoRA Artifact

deathmachine069 Final Proposal

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Abstract

LLMs sometimes, acquire unwanted behavior, by generalizing patterns that are not explicitly present in training data. The main research question that is to be discussed in this research project is, How does LoRA-rank (Hu et al., 2022) affect the presence Inductive Backdoor behavior?

We plan on implementing the US Presidents experiment for demonstrating inductive-backdoor behavior, from the Betley et al. (2025) paper. The primary model will be Qwen2.5-3B (Yang et al., 2025). We will try mechanistic analysis done by Nief et al. (2026), We will try to test, by doing SVD on the LoRA matrix, if some of the directions are enough to reproduce the inductive-backdoor behavior. We will also try, dynamic LoRA grafting, enable or disable the adapter at the token span encoding the president number, allowing comparison with random padding tokens. This project aims to figure out if inductive backdoor generalization is a consequence of LoRA.

1 Introduction

LLM generalization sometimes leads to weird behaviors, behaviors that were not present in the fine-tuning data. Inductive backdoors, introduced by Betley et al., in this the LLM recognizes a held-out trigger and is able to replicate behavior through a generalized rule from previous shown triggers.

The US presidents experiment from the Betley et al paper, were done to test if a LLM could learn a hidden general rule that connected varied triggers to behaviors that they were not paired explicitly during the fine-tuning. This was done by generating a sequence of numbers that were mapped externally to US presidents, and tests were done to see if the behavior is replicated when unseen numbers in the sequence are shown to the model.

We intend on reproducing these effects on the Qwen2.5-3B, after fine-tuning at different LoRA ranks, singular-value analysis, dynamic LoRA

grafting, and activation patching. To make the task more computationally feasible for our current scale, we have limited ourselves to fine-tuning on a smaller number of LoRA ranks, use a smaller model than the ones used, a small generated dataset, and less adapter locations. The main research question that is to be answered here is, Does LoRa-rank affect the measurable unseen weird generalization and how does the behavior and its occurrence change with the LoRA-rank being modified?

2 Background

LoRa, is a fine-tuning technique, that keeps the pre-trained weights of an LLM frozen and learns a lo-rank update to a few selected weight matrices. The thing that makes LoRA favorable for fine-tuning under limited compute might also lead to different kinds of memorization, generalization, or unintended behavior.

$$W = W_0 + \frac{\alpha}{r}BA,$$

Nief et al. Paper talks about an inverted-U relationship between subliminal learning occurring in LLMs and LoRA rank. An inductive backdoor, Betley et al, is different from a conventional backdoor, because on the generalizations it has to make over held-out data. Successful generalization would suggest that the model learned a general rule that trigger had some underlying structure that linked it to a pre-trained semantic entity, and not just memorizing the training examples.

Recent developments suggest multiple explanations for why these weird generalizations occur after fine-tuning. The paper by Wang et al, says that some conditional behaviors can be explained by a constant steering vector that shifts the model toward a general concept. There have also been findings by Soligo et al. that suggest independently fine-tuned models can converge to similar activation directions associated with a learned behavior. Blank et al. also talks about hidden

behavioural transfer being understood as steering-vector distillation. Nief et al. also showed that LoRA-based effects that have a relation to subliminal learning can also depend on fine-tuning and evaluation context, token positions, and particular low-rank directions.

These findings make us question if inductive-backdoor behavior can be explained by a low-dimensional steering direction, a LoRA effect, or is a behavior that consistently emerges across different training runs.

3 Project Goals and Research Questions

The primary research question and two secondary checks are:

- We want to test how different LoRA ranks affect the unseen trigger codes on fixed training examples on the decided base model with layer location for the adapter being fixed.
- does the structure of the trigger matter? does the inductive backdoor behavior get modified when the President code is removed, moved, or randomized.
- We want to check if the behaviour is controlled by activation in one single direction, in successful run that show inductive behavior or a if there a complex mechanism that is being learnt.

The main hypothesis is that at least one rank will show above-chance generalization on held-out trigger values, but the strength and stability of the effect will vary across ranks and seeds. Nief et al. talk about an inverted-U possibility, but this project will not assume this, we will observe if the curve is increasing, decreasing, non-monotonic, or indistinguishable from noise.

4 Dataset

A modified version of the experiment on US presidents from Betley et al, in which we will use programmatically generated proxies, will be created. The inputs for the task are an eight-digit code and an index position. It is required to state the name of the US president according to the given example. In the training dataset, there will be eight indices where about 30 random codes will be generated for each (240 examples). Also, there will be four indices where about 30 codes will be generated for each of them, but they will not be used during LoRA training.

The following four conditions will be included in the evaluation set. They are as follows. Seen indices, unseen indices, codes with the index positioned incorrectly, prompt without any code. The evaluation metric will be exact match accuracy rate after light normalization. We choose president names because they are already semantic entities recognized by the pretrained model, while the codes are unknown to the model.

5 Methodology

5.1 Model and LoRA setup

We will choose Qwen2.5-3B-Instruct to be our base model, this is not the same as the models used in similar experiments in the papers, but we have chosen it due the compute constraints we have.

For each primary run, we will attach a LoRA adapter placing a low-rank adapter only to the MLP down-projection of one fixed transformer layer. we will do different runs varying this on early, middle and late layers, for example layers 6, 18, and 30 of the 36-layer model. rather than to every attention and MLP matrix, this is also due to lower compute constraints. If no behavior emerges, we will place adapters on the MLP down-projections of three layers (early, middle, late) for three-layer condition.

For Training we will use short sequences (around 128 tokens) and batch size 1. For sweep we will use two seeds per rank, with a third seed reserved for the strongest condition.

5.2 Behavioral evaluation

On each run, we will measure exact-match accuracy on seen and unseen indices, generalization gap and trigger specificity.

Specificity will be estimated by comparing the normal coded prompt with matched prompts where the code is absent, the index is moved, or the digits are randomly permuted.

Since context mismatch is central to Nief et al, we will also evaluate a matched default Qwen system prompt againsta an empty/changed system prompt. A non-LoRA baseline will establish the base model’s capabilities.

5.3 Lightweight activation analysis

If some run shows reliable unseen-trigger generalization, we will run/reproduce these three tests in Nief et al:

- Dynamic LoRA grafting will turn the adapter

on only at the trigger, shared system-prompt, chat-template, or random-padding token positions, and separately in early versus late layers.

- Compute SVD of the learned BA update and replace it with its first singular component. This is to test whether the behaviour will still be the same in a rank-construction.
- We will patch the MLP down-projection output from a successful donor prompt into a matched baseline or wrong-trigger prompt for some examples. We will also compute a mean difference activation direction and compare it with the LoRA output direction.

5.4 Fallback

The project does not remain inconclusive if the main experiment does not produce reliable ‘held-out-index’ generalization. First step is to determine if there are any implementation errors, we can check this by seeing prompt formatting, training duration, LoRA rank, and layer placement. By checking for these results, we can make fair deductions of what happened, for instance If the model learns the seen-index task but fails on held-out indices, then the chosen LoRA intervention can learn the trigger task but does not induce the expected inductive generalization.

In such cases, we will investigate alternate mechanisms. We have outlined few below:

- We can test whether the learned behaviour can be explained by an unconditional or index-specific steering direction rather than a genuinely conditional trigger mechanism. We will measure mean activation differences and test whether injecting these directions into no-code or invalid-trigger prompts changes the model’s president-name predictions
- We can compare LoRA updates and activation directions across random seeds, ranks, and layer locations to test whether independently trained adapters converge to similar directions.
- It might do good to also examine rank dependence, system-prompt matching, token-level localization, rank-one SVD reconstructions, and activation patching. Finally, we can compare the learned LoRA effect with a separately constructed activation steering vector to test whether the behaviour is better characterized as low-dimensional steering than as a discrete

backdoor.

These follow up experiments are motivated by recent studies of out-of-context reasoning, convergent fine-tuning representations, LoRA-dependent behaviour, and steering-vector distillation.

6 Expected Outcomes and Limitations

The expected outcome is a small reproducible benchmark showing whether rank changes the probability or reliability of unseen-trigger generalization in this controlled setting. A positive result would support the claim that low-rank parameterization is sufficient for inductive backdoor like behavior and would motivate the activation-direction check. A negative result would still be useful: it would show that the phenomenon may require a larger model, a different trigger construction, or more training signal than this setup provides.

The limits of this project are , Qwen2.5-3B is not a frontier model, the synthetic task is not a safety benchmark, and the experiment is not exactly full fine-tuning. The final conclusion will only be about rank sensitivity and mechanism in a 3B-class model, and cannot be used to state whether LoRA is universally responsible for inductive backdoors.

References

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